

ORCHIDEE WP2 – SARI – Indicator specification form for absolute incidence

Version 1.1 of 29/09/2025

This indicator specification form is part of the epidemiological surveillance system for Severe Acute Respiratory Infections (SARI). It compiles all epidemiological definitions required for the standardised calculation of the absolute incidence indicator, in accordance with the three case definitions adopted within the framework of the Orchidée project.

	Epidemiological definition
Context	The definition of Severe Acute Respiratory Infections (SARI) cases and the related indicators to be produced by Santé publique France are based on the ECDC's SARI reporting protocol within the framework of the European Surveillance System (TESSy)¹. ECDC defines SARI cases as any patient with an acute respiratory infection requiring hospitalization, accompanied by a fever exceeding 38°C and a cough within a span of 10 days: « - Patient with an acute respiratory infection that requires hospitalisation AND: - Has a history of fever or measured fever of $\geq 38\text{ C}^\circ$ AND - cough AND - with an onset within the last 10 days. » In the context of the Orchidée project, a proxy definition is defined through ICD-10 codes for acute lower respiratory infections (see table 1 for the exhaustive list) J09* to J22* + U07.1* (except U07.13) <i>*Including sub-codes</i>
Objective	To monitor trends of SARI and their impact on hospitalizations and in-hospital mortality to guide public health decisions.
Inpatient stay definition	A hospital stay is defined as the time period between a patient's admission and definitive discharge from an hospital (Fig. 1). More precisely : 1. It includes only stays in MCO (Médecine- Chirurgie-Obstétrique) units (see table 2 for the list of MCO units, as per the PMSI definition). 2. It begins with the first hospital contact (including emergency units). 3. It ends with the definitive discharge date from an MCO unit. A definitive discharge is a discharge that is not followed by a readmission within 24 hours. 4. If a hospital discharge and a subsequent MCO admission are separated by less than 24 hours, they are considered part of the same hospital stay.

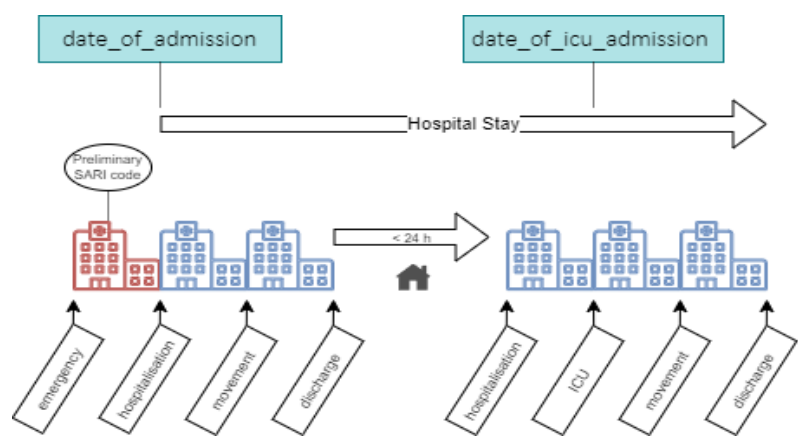


Figure 1: Example of a SARI hospital stay identified by a preliminary diagnosis (during emergency admission).
The hospital stay includes multiple venues (due to a re admission in less than 24h)

¹ ECDC. The European Surveillance System (TESSy) Severe Acute Respiratory Infections (SARI) Reporting Protocol Version 3.8 du 14 august 2023 [Internet]. Link: <https://www.ecdc.europa.eu/en/publications-data/severe-acute-respiratory-infections-sari-reporting-protocol>. Consulted on November 2024, the 11th.

Case definitions (CD)	Three different case definitions were developed:		
	<u>Reactive SARI CD</u> corresponds to (excluding PMSI) that can be identified within the hospital information system. - It includes (and mainly corresponds to) ICD-10 codes recorded during emergency visits. - It excludes PMSI ICD-10 diagnosis	<u>Specific SARI CD</u> corresponds to any SARI ICD-10 diagnosis as the principal diagnosis (DP) from the health data warehouses (EDS) of the Hospital Information System (PMSI).	<u>Sensitive SARI CD</u> corresponds to any SARI ICD-10 diagnosis consolidated as the principal (DP) or associated diagnosis (DAS) from the health data warehouses (EDS) of the Hospital Information System (PMSI).
Data source	Hospital Health Data Warehouse <u>except PMSI</u>	French Hospital Information Program (PMSI)	French Hospital Information Program (PMSI)
Event of interest	Three different events of interest are considered:		
	<u>SARI hospital admission:</u> New inpatient stay fulfilling one of the SARI CD	<u>SARI ICU admission:</u> SARI hospital admission that includes at least one admission to an Intensive Care Unit (ICU)	<u>SARI hospital death:</u> SARI hospital admission where the patient dies during the inpatient stay
Target populations	Hospital admission: All new inpatient stays (see table 2 for the list of MCO units, ATIH definition)	ICU admission: All hospital admission that includes at least one stay in an ICU (see table 2 for the list of ICU, ATIH definition)	Hospital death: All hospital admissions during which the patient dies during their inpatient stay
Time reference:	Week of hospital admission	Week of first ICU admission	Week of inpatient death
	<u>Week</u> is defined as the epidemiological week: Monday 00:00 to Sunday 23:59; ISO 8601 norms: the first week of the year is the week with a majority (four or more) of its days in January.		
Forms of result	For each <u>case definition</u> and for each <u>event of interest</u> : Incidence = Number of events by time reference Stratification: - By age group (< 1 month ; 1 to 2 months ; 3 to 5 months ; 6 to 11 months ; 12-23 months ; 24 months to 4 years ; 5 to 14 years ; 15 to 29 years ; 30 to 64 years ; 65 to 74 years ; 75 to 79 years ; ≥ 80 years ; unknown) - By biological sex: male, female, other		
Production frequency	Weekly To send to Santé publique France before each Tuesday 12 pm (Paris time)		

Table 1 : List of syndromic groups, corresponding ICD-10 codes and conditions used for SARI case definition in the Orchidée projet. Source: atih.sante.fr.

Syndromic groups	ICD-10 codes	Corresponding condition
Influenza	J09	Influenza due to identified zoonotic or pandemic influenza virus
	J10	Influenza due to identified seasonal influenza virus
	J10.0	Influenza with pneumonia, influenza virus identified
	J10.1	Influenza with other respiratory manifestations, influenza virus identified
	J10.8	Influenza with other manifestations, influenza virus identified
	J11	Influenza, virus not identified
	J11.0	Influenza with pneumonia, virus not identified
	J11.1	Influenza with other respiratory manifestations, virus not identified
	J11.8	Influenza with other manifestations, virus not identified
Pneumonia	J12	Viral pneumonia, not elsewhere classified
	J12.0	Adenoviral pneumonia
	J12.1	Pneumonia due to respiratory syncytial virus (RSV)
	J12.2	Pneumonia due to parainfluenza viruses
	J12.8	Other viral pneumonia
	J12.9	Viral pneumonia, unspecified
	J13	Pneumonia due to <i>Streptococcus pneumoniae</i>
	J14	Pneumonia due to <i>Haemophilus influenzae</i>
	J15	Bacterial pneumonia, not elsewhere classified
	J15.0	Pneumonia due to <i>Klebsiella pneumoniae</i>
	J15.1	Pneumonia due to <i>Pseudomonas</i>
	J15.2	Pneumonia due to staphylococci
	J15.3	Pneumonia due to group B streptococci
	J15.4	Pneumonia due to other streptococci
	J15.5	Pneumonia due to <i>Escherichia coli</i>
	J15.6	Pneumonia due to other Gram-negative aerobic bacteria
	J15.7	Pneumonia due to <i>Mycoplasma pneumoniae</i>
	J15.8	Other bacterial pneumonia
	J15.9	Bacterial pneumonia, unspecified
	J16	Pneumonia due to other infectious organisms, not elsewhere classified
	J16.0	Pneumonia due to <i>Chlamydia</i>
	J16.8	Pneumonia due to other infectious organisms
	J17	Pneumonia in diseases classified elsewhere
	J17.0	Pneumonia in bacterial diseases classified elsewhere
	J17.1	Pneumonia in viral diseases classified elsewhere
	J17.2	Pneumonia in mycoses
	J17.3	Pneumonia in parasitic diseases
	J17.8	Pneumonia in other diseases classified elsewhere
	J18	Pneumonia, organism unspecified
	J18.0	Bronchopneumonia, unspecified
	J18.1	Lobar pneumonia, unspecified
	J18.2	Hypostatic pneumonia, unspecified
	J18.8	Other pneumonia, organism unspecified
	J18.9	Pneumonia, unspecified
Other acute lower respiratory infections	J20	Acute bronchitis
	J20.0	Acute bronchitis due to <i>Mycoplasma pneumoniae</i>
	J20.1	Acute bronchitis due to <i>Haemophilus influenzae</i>
	J20.2	Acute bronchitis due to streptococci
	J20.3	Acute bronchitis due to Coxsackie virus
	J20.4	Acute bronchitis due to parainfluenza viruses

	J20.5	Acute bronchitis due to respiratory syncytial virus (RSV)
	J20.6	Acute bronchitis due to rhinovirus
	J20.7	Acute bronchitis due to ECHO virus
	J20.8	Acute bronchitis due to other specified organisms
	J20.9	Acute bronchitis, unspecified
	J21	Acute bronchiolitis
	J21.0	Acute bronchiolitis due to respiratory syncytial virus (RSV)
	J21.8	Acute bronchiolitis due to other specified organisms
	J21.9	Acute bronchiolitis, unspecified
	J22	Unspecified acute lower respiratory infection
Provisional assignment of new diseases of uncertain etiology or emergency use	U07.1	COVID-19, virus identified
	U07.10	COVID-19, respiratory form, virus identified
	U07.11	COVID-19, respiratory form, virus not identified
	U07.12	Asymptomatic carrier of SARS-CoV-2
	U07.14	COVID-19, other clinical forms, virus identified
	U07.15	COVID-19, other clinical forms, virus not identified

Table 2 : List of Medical and intensive care units* (PMSI-MCO) used for SARI surveillance in the Orchidée project. Source: *atih.sante.fr*

Code Autorisation	Classification de l'unité médicale fonctionnelle (après réforme des autorisations)
01A	Réanimation adulte hors grands brûlés
01B	Réanimation adulte grands brûlés
02A	Soins intensifs en cardiologie = USIC
02B	Autres soins intensifs (hors UNV, USIC, néonatalogie)
02C	Soins intensifs polyvalents adultes
02D	Soins intensifs polyvalents dérogatoires adultes
02E	Soins intensifs en cardiologie = USIC adultes
02F	Soins intensifs neurovasculaires (USINV) adultes
02H	Soins intensifs d'hématologie (USIH) adultes
02I	Soins intensifs de spécialité dont Soins intensifs de néphrologie Soins intensifs respiratoires Soins intensifs d'hépto-gastro-entérologie
03A	Soins surveillance continue adulte hors grands brûlés
03B	Soins surveillance continue adulte grands brûlés
04	Néonatalogie sans SI
05	Soins intensifs en néonatalogie
06	Réanimation néonatale
07A	UHCD structures des urgences générales
07B	UHCD structures des urgences pédiatriques
07C	UHCD des antennes de médecine d'urgence générale
08	Soins et accompagnement des malades en phase terminale (unité de soins palliatifs)
13A	Réanimation pédiatrique hors grand brûlés
13B	Réanimation pédiatrique grands brûlés
13G	Réanimation pédiatrique de recours hors grand brûlés
14A	Soins surveillance continue pédiatrique hors grands brûlés
14B	Soins surveillance continue pédiatrique grands brûlés
15C	Soins intensifs polyvalents pédiatriques
15D	Soins intensifs polyvalents dérogatoires pédiatriques
15E	Soins intensifs pédiatriques de cardiologie
15G	Soins intensifs polyvalents pédiatriques de recours
15H	Soins intensifs d'hématologie (USIH) pédiatriques
15I	Soins intensifs pédiatriques de spécialité dont Soins intensifs de néphrologie Soins intensifs respiratoires Soins intensifs d'hépto-gastro-entérologie
16	Unité d'hématologie équipée d'un système de traitement de l'air
17	UNV hors SI
18	Soins intensifs en UNV
19	Unité d'addictologie de recours et de référence
20	Traitement des grands brûlés (hors réanimation et surveillance continue grands brûlés)
21	Hémodialyse en centre pour adulte
22	Hémodialyse en centre pour enfant
23	Hémodialyse en unité médicalisée
26	Unité hospitalière sécurisée interrégionale UHSI

27	Médecine gériatrique
28	Médecine pédiatrique (hors unité d'oncologie médicale, spécialisée ou non)
29	Autres spécialités médicales adultes (non classées ailleurs) ou unité de médecine indifférenciée
30	Unité pour la prise en charge d'IOA
34	Hémodialyse en unité d'autodialyse
35	Hémodialyse à domicile
36	Dialyse péritonéale à domicile
37	Unité de dialyse saisonnière
40	Unité d'hospitalisation d'oncologie médicale
41	Unité d'hospitalisation d'oncologie chirurgicale
42	Unité de radiothérapie ambulatoire
43	Unité de chimiothérapie ambulatoire
50	Chirurgie cardiaque
51	Neurochirurgie
52	Autre chirurgie pédiatrique (ou chirurgie indifférenciée pédiatrique)
53	Autre chirurgie adulte (ou chirurgie indifférenciée adulte)
60	Centre pour la prise en charge des valves aortiques
61	Unité de prise en charge de la douleur chronique
70	Gynécologie seule
71	Obstétrique seule
73	Gynécologie obstétrique indifférenciée
77	Plus de 2 types d'UM différents dans la même zone d'UM (type_RUM_2)
88	Fin d'autorisation de l'unité

* In grey, the intensive care units